

摘要

引言

2012年希格斯粒子的发现不仅是粒子物理领域重要的里程碑之一，同时也开启了一个研究新物理的全新窗口，这是因为希格斯粒子与粒子物理领域现有的许多重大疑难问题密切相关。例如希格斯自相互作用机制直接关系着真空相变形式、宇宙中物质反物质不对称性的形成，暗物质粒子的质量产生机制是否也跟希格斯粒子相关等等，这些问题都是现有的标准模型理论不能解释的，必然要求存在着超越标准模型的新理论。

许多超出标准模型的新理论所预测的希格斯粒子的性质与标准模型的预测有所不同，因此通过深入研究希格斯粒子的性质不仅可以直接检验标准模型理论，也可以探索新物理的迹象，从而可能回答许多粒子物理学的关键科学问题。例如Higgs横向动量分布可以用于检测微扰QCD的计算，也对Higgs的拉氏量的结构非常敏感；Higgs的快度分布对质子的部分子分布函数（PDFs）非常敏感；jet相关的观测量对理论模型非常敏感等等。实验中如观测到任何与标准模型的偏差，都将直接指向超越标准模型的贡献。CMS和ATLAS合作组在不同Higgs衰变道中对这些问题进行了深入研究 [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]。

本论文通过希格斯玻色子衰变到四轻子过程研究了希格斯粒子的性质。基于CMS探测器收集的质心系能量为13 TeV的质子质子对撞数据，测量了希格斯至四轻子衰变的基准截面，包括积分截面和微分截面。这些测量使用了CMS第二期运行获取的全部数据，即质心系能量为13 TeV、积分亮度为 137.1fb^{-1} 的数据，在测量中希格斯质量取值为125.09 GeV。为了使所测量结果不依赖于希格斯的理论预测，论文基于末态轻子运动学信息和事例拓扑结构定义了基准相空间，所有的截面测量都限定于基准相空间。在质心系能量为13 TeV情况下，积分基准截面测量值为 $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})\text{fb}$ 。另外针对多个对希格斯玻色子产生机制敏感的运动学观测量，包括快度、四轻子体系的横动量、相关的多重喷注以及领头喷注的横动量等进行了基准微分截面测量。测量结果直接检验了标准模型希格斯粒子的产生机制，以及TeV能标上的微扰QCD计算，对于检验标准模型中关于希格斯粒子的预测至关重要，同时可以用于约束许多超标准模型理论的相空间。在误差范围内，积分截面和微分截面结果均与标准模型理论预言一致。

理论背景

此次测量是为精确检验粒子物理标准模型（SM）。经过长时间的发展，标准模型可以精确描述物质及物质间的相互作用（电磁力，弱力，强力不包括引力）。这些基本作用力的细节描述见??。标准模型对基本相互作用的描述由数学的对称法则组成，它提供的数学描述只能适用于质量为0的粒子。Higgs机制的引入，使得标准模型更加普世，解决了标准模型只能描述0质量粒子的问题。在Higgs机制中，粒子通过电弱对称性破缺获得质量。Higgs粒子的存在就是Higgs机制的直接证明 [14, 15, 16, 17, 18, 19]。标准模型是用 $SU(3)_C \otimes SU(2)_L \otimes U(1)_Y$ 规范对称群描述的量子场论。它的公式化源自不同时期的不同科学家跨越几十年的相互协作。1961年，Glashow提出电磁力和弱力有相似的结构，可以统一用

$SU(2)_L \otimes U(1)_Y$ 对称群解释。Brout, Englert和 Higgs 提出无质量的基本粒子可以通过自发对称性破缺以及相关的另一个基本粒子Higgs 玻色子获得质量。之后, Salam和Weinberg提出Glashow的 $SU(2)_L \otimes U(1)_Y$ 模型中L表示左手征态费米子场, Y是弱超荷, 与弱同位旋和电荷相关, 至此完成了标准模型中的电弱统一。如 ?? 中的描述, Glashow对电磁力和弱力的统一描述只适用于无质量粒子。然而规范场要求粒子必须拥有质量, 以描述弱相互作用以及和实验结果相一致。Higgs机制被提出作为标准模型的一部分, 同时引入一个新的标量场, 同时这个标量场的变换满足规范协变性和重整化。自发对称破缺的公式化概念以及费米子和玻色子 (W, Z) 的质量起源详见 [14, 15, 16, 17, 18, 19]。关于Higgs粒子的细节描述见 ??。可以说Higgs玻色子是粒子物理标准模型的基石, 它是2012年在LHC上被CMS和ATLAS两个实验组同时发现的。

LHC中质子-质子对撞现象学

LHC是同类加速器物理实验中最大的实验装置, 建造它的目的是为了发现希格斯粒子并精确检验粒子物理标准模型。质子由三个价夸克 (uud) 通过交换胶子的强相互作用束缚在一起。夸克通过一些方法在质子内部相互作用。一个夸克与另外两个夸克彼此相互作用, 或者通过湮灭过程产生胶子, 胶子进一步产生夸克反夸克对。结果导致除了价夸克外, 质子内部还有海夸克和胶子 (统称为部分子)。在对撞实验中, 大的动量转移可能发生在对撞的部分子之间, 即硬散射。一些共振态可能在这个过程中产生比如Higgs粒子。一个过程发生的可能性被定义为该过程的截面。LHC中质子-质子对撞的不同过程的产生截面见图 ??。没有参与硬散射的部分子通常会直接向前飞走后参与低动量过程。低动量过程事例被称为underlying事例。如果在一次质子-质子对撞事例中有超过一个相互作用发生, 称为多部分子相互作用 (MPI)。被加速的带电部分子可能会向外辐射光子和胶子, 被称为韧致辐射。如果这些辐射可以关联到一个入射部分子, 这辐射成为初态辐射。如果关联到一个出射部分子, 成为末态辐射。之后夸克和胶子通过强子化过程 (相互束缚形成色单态) 组成一些强子。这些强子经常表现为喷注, 称为jet, 如 ?? 中的描述。在硬散射中, 表示一个过程发生可能性的截面可以通过Factorization (μ_F) 定理计算。这个定理使用Factorization因子帮助我们将截面分解为硬、软过程。关于截面的数学描述见 ??。

希格斯粒子的产生及衰变

本文主要涉及5个不同的Higgs产生模式。在LHC中最主要的Higgs产生模式是胶子-胶子融合。胶子通过虚夸克圈图与Higgs耦合。因为耦合直接与费米子质量成正比, 所以top夸克圈图在这里占主导。VBF是第二重要的Higgs产生模式, 是由两个W或者Z玻色子融合产生Higgs粒子。VBF过程的实验信号因为带有两个向前的高能jet的特征, 所以可以与胶子融合信号有很好的区分。同时这个特征也用来压低本底。在与W/Z联合产生模式中, Higgs粒子由高能W或Z玻色子产生, 这个W或Z玻色子由两个费米子融合产生。该过程在LEP实验中是最主要的Higgs产生过程, Tevatron实验中是次级产生过程。在LEP正负电子对撞实验中, 该过程产生了很多W和Z玻色子。在强子对撞机中, Tevatron使用质子反质子对撞, 这与LHC更相似。更小产生截面的Higgs产生过程是由两个胶子碰撞产生top和反-top夸克对, 之后 $t\bar{t}$ 融合产生Higgs。

标准模型125 GeV Higgs粒子的寿命大约是 1.6×10^{-22} s, 它可以衰变到所有标准模型有质量粒子 (在壳重粒子的可能性更大)。Higgs玻色子衰变到不同粒子的

分支比如图 ??所示 [20]。 $H \rightarrow b\bar{b}$ ($m_H \approx 125$ GeV) 是分支比最大的衰变模式。但是该衰变道的QCD背景比信号高几个数量级, 所以该衰变道的信噪比最低。然而, CMS和ATLAS合作组以及一些其他实验比如 [21], [22] 和 [23] (CDF和D0) 仍然利用了这个衰变道。B夸克在探测器中也很难重建。当它们的在探测器中衰变时, 它们的碎片产生很多jet和强子。我们很难完美的重建这些jet。正是因为如此, 我们必须开发其他衰变道。Higgs衰变到双光子是非常重要的衰变道, 因为它有两个标志性的高能光子和好的质量分辨率。因为Higgs玻色子的质量在125GeV, Higgs玻色子也可以衰变到两个矢量玻色子。这里面一个或者两个玻色子可能是不在壳的。WW衰变可以进一步衰变到中微子, 但是中微子在探测器中不可能完全重建。 $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ 衰变道尽管衰变分支比很低, 但是这些轻子是高能轻子, 可以在探测器中得到完全重建, 并且末态不变质量分辨率很高 [20], 被视为黄金衰变道。本论文中对Higgs粒子性质的测量正是基于 $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ 衰变道。

实验背景

Tevatron、LEP高能物理实验的精确测量证明了标准模型的正确性, 但是仍然没有观测到Higgs粒子。2010年, 大型强子对撞机正式运行, 将对标准模型做更多的检验, 以及验证之前的实验结果。LHC安装在欧洲核子研究中心 (CERN), 坐落在瑞士-法国边境, 日内瓦近郊。它是粒子物理同类实验中能量最高的实验装置。它有足够的能将质子束流加速到7TeV, 设计的亮度峰值为 $\mathcal{L} = 10^{34} \text{cm}^2 \text{s}^{-1}$ 。它是人类有史以来建造过的最先进、最复杂和最大的实验装置, 由来自超过100个国家的10000名工程技术人员以及科学家共同设计建造。LHC的建造是从1998年开始, 一直到2008年, 是在地下75-150米深的3.8米宽、27千米长的隧道里。该隧道在1983-1988年曾经用来安装LEP对撞机。在LHC中两个方向相反的质子束流运行在高真空的管道中以避免束流和空气分子的额外碰撞。

2012年, 在LHC上运行的两个实验CMS和ATLAS联合宣布发现了一个新的重标量玻色子, Higgs玻色子 [24, 25]。基于这个发现, 物理学家希望通过精确测量Higgs粒子的性质, 以进一步检验标准模型。Higgs粒子的质量是Higgs机制中的唯一自由参数, 它的测量结果是 $m_H = 125.09 \pm 0.21(\text{stat.}) \pm 0.11(\text{syst.})$ GeV [26]。

在运行第一阶段成功取数后, 大型强子对撞机关闭了两年。在2013年至2014年间LHC加速器设备和探测器进行了重大升级。2015年, LHC重启, 质心系能量达到13TeV (接近其设计值14TeV)。2015年CMS采集了质心能量为13TeV、积分亮度为 4.2fb^{-1} 的数据。随后在2016、2017和2018年间, CMS分别累积了 41.0fb^{-1} , 49.8fb^{-1} , 67.9fb^{-1} 的数据。在第二阶段运行后, LHC目前正在经历第二阶段的长期关闭。在此期间, 多个升级改造正在进行, 为后期LHC高亮度运行 (HL-LHC) 做准备。但是, 这方面的主要升级将在第三次关闭期间进行 (即大型强子对撞机运行第3阶段之后)。其一是用Linac4替换Linac2。这个长度约为90米, 耗时约10年建成, 用于加速负氢离子 (H^-), 使粒子束能量达到160 MeV, 并使其强度翻倍。此外, 在第二阶段长停机期间, LHC上各个实验组, 如CMS, ATLAS, ALICE和LHCb, 也在进行相应的升级改造。ATLAS正在构建与触发系统集成的缪子小轮。为了应对高堆积 (pileup), LAr电子设备也正在升级。CMS将强子量能器桶部和端盖的光电探测器 (HPD) 更改为硅光电倍增管 (SiPM)。此外, 用铝代替不锈钢制作束流管。由于辐射造成了强子量能器端盖的损伤, 闪烁条也将更换。为了准备高亮度运行 (HL-LHC), CSC缪子室正在升级改造, 装备新的前端 (FE) 电子学器件。ALICE正在升级内部径迹系统 (ITS),

时间投影室 (TPS)，中央触发处理器，数据采集 (DAQ) 高级别触发 (HLT)，缪子前向径迹探测器 (MFT) 等。LHCb 将其数据采集从 1 MHz 提高到 40 MHz，将用完全基于软件的触发方式替代前端 (FE) 电子学器件以及使用全新的计算基础架构。

紧凑型缪子探测器

本文使用的数据来自于 CMS 探测器。有关 CMS 探测器的构造、内部结构以及 CMS 探测器如何重建事件的说明，请参见章节 ??。大型强子对撞机基于最新技术，配备了最先进的实验设施，为粒子物理实验研究开辟了新领域。为了确保实验结果的有效性，LHC 有两个主要的目标基本一致的通用型实验进行交叉验证，CMS 是其中之一。它们具有相同的物理目标，但探测器的设计和技术不同。CMS 能够在更广范围内为 μ 识别和动量提供良好的分辨率，其动量在 1 TeV 上仍有极致的精确度。CMS 探测器具有良好的带电粒子内部径迹重建效率和分辨率。它具有高效的触发系统，并能更好地识别 τ 和 b 喷注。此外，双电子、双光子、双喷注以及缺失横向能量 (MET) 具有优异的质量和能量分辨率。这使其能够研究标准模型物理 (包括希格斯物理)，额外维度和暗物质等新物理，覆盖 100 GeV 到 TeV 能量尺度的物理 [27]。CMS 探测器是圆柱形，长度约为 28 m，外半径大约 7 m，总高度约为 15 m，总重量约为 14000 吨。

CMS 从对撞点径向向外分别为内部径迹探测器 (ITS)、电磁量能器 (ECAL)、强子量能器 (HCAL) 和缪子室。这些子探测器处在大型磁场内。CMS 的内部径迹探测器具有较大的径迹多样性，因为它最接近发生对撞的束流管。电磁量能器由钨酸铅晶体构成，涵盖了赙快度小于 3.0 的范围。在电磁量能器后是由黄铜制成的强子量能器。CMS 探测器的最外层是缪子系统。

LHC 每秒发生 4000 万次碰撞 (40 MHz)，但是并不是所有的碰撞都是感兴趣的研究物理事例。此外，数据采集系统 (DAQ) 不能有效地高速率读取每个事件。存储碰撞中产生的无趣事件也将消耗大量的存储。因此，需要触发系统，用以减少在碰撞期间记录事件的速率。触发系统将选择出仅对物理目标至关重要的事件。CMS 的触发系统是一个复杂的两级系统，包括第一层触发 (L1) 和高级别触发 (HLT)。

希格斯基准截面的测量策略

标准模型的希格斯玻色子在 LHC 上有五个主要的产生模式 (ggH, VBF, WH, ZH 和 ttH)，由 POWHEG V2 产生子产生这些信号样本。WH 和 ZH 是通过 POWHEG 的 MINLO HVJ 扩展进行模拟 [28]。希格斯到四轻子衰变过程由 JHUGEN 产生子模拟 [29]。产生 WH, ZH 和 ttH 样本时，允许希格斯衰变到 $H \rightarrow ZZ \rightarrow 2\ell 2X$ ，相应地，四轻子事件的两个轻子有可能分别从关联的 W, Z 玻色子或顶夸克的衰变中产生。使用 PYTHIA8.209 模拟部分子的簇射，并且在所有情况下，簇射的匹配标准是先允许所有能量范围的量子色动力学 QCD 发散，然后在 POWHEG 等级下进行判选。所有样本均使用默认的 NNPDF 3.1 NLO 部分子分布函数 (PDF) [30]。这些信号样本截面信息包含在 [12]。这些不同产生模式的样本还用于评估测量结果对模型的依赖性，如表 ?? 所示。关于评估模型依赖性的方法的详细信息将在 ?? 节中进行描述。ggH 产生模式的 POWHEG 和 NNLOPS 样本被用于标准模型计算的理论预测，以及与测量结果进行比较。本底估计中使用 POWHEG V2 和 PYTHIA8 交互产生次领头阶 (NLO) 精度的主导过程 $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ 。gg $\rightarrow ZZ$ 过程使用 MCFM 进行领头阶精度 (LO) 的模拟。有关模拟和数据样本的更多详细信息，请参见 ?? and

??部分。希格斯截面的测量，包括积分截面和微分截面，是在基准相空间中进行的，重建选择条件与基准空间进行了匹配。对于希格斯事例的挑选，首先从选定的轻子中构建四轻子候选者，轻子的选择条件包括顶点约束 $SIP_{3D} < 4$ (SIP_{3D} 是个重要的影响参数，其值是将影响参数除以相关的误差)，以及减去其中末态辐射光子贡献的孤立性选择 (isolation cut)。Z玻色子是由相反电荷的相同味道的轻子(e^+e^- , $\mu^+\mu^-$)构成，并且要求其不变质量在12GeV到120GeV范围内。然后从不重叠的Z玻色子候选对象构建ZZ候选对象。其中不变质量更接近于标准模型Z玻色子的被标记为 Z_1 ，另一个为 Z_2 。信号区间由至少要包含一个ZZ候选者的事例构成，更多细节参见章节??。

$pp \rightarrow H \rightarrow 4\ell$ 的积分和微分基准截面的测量由信号和本底参数化的函数对包含四轻子质量分布的数据进行最大似然拟合得到，并直接从中提取基准截面。系统误差以干扰参数的形式包括在内，并在拟合过程中得到有效整合。使用渐近方法 [31]和基于最大似然比 [32]的检验统计量获得结果。从观察到的分布中进行探测器效应的修正，即解谱过程 [33]和 [34]。在微分截面测量中，将有限的效率和分辨率效果编码到探测器响应矩阵中，该矩阵描述事件如何从基准级别的既定观察量单元映射到重建级别的既定单元。该矩阵是对角线主导的，只有涉及喷注的观测量中含有较多的非对角线元素。

系统误差的主要来源是实验误差，包含轻子鉴别效率和积分亮度测量不确定性，理论误差来源是次要的。为了评估测量的模型依赖性，使用不同的响应矩阵重复重复测量过程，这些响应矩阵是通过在其实验约束范围内更改每个标准模型产生模式的相对部分而创建的。相对于实验系统误差，这个不确定性导致的误差很小，是可以忽略的。

结果和展望

为了使所有测量结果独立于希格斯的产生方式，基于轻子运动学和事件拓扑结构定义基准相空间，所有的截面测量都取自基准相空间。在质心系能量13TeV下，测量的积分基准截面为 $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})$ fb。基准微分截面则测量了快度、四轻子体系的横动量、相关的多重喷注以及领头喷注的横动量的分布。微分截面在这些观测量的各个单元内进行测量。对于每个运动学或微分观测量，优化每个单元边界的选择过程，以使观测量的每个单元具有相似的预期测量横截面。微分横截面测量应在观测量的每个单元上的重建层次上减去背景。对于所有微分单元而言，本底的质量分布是平滑的，没有共振结构的。对于不可区分和可区分的背景，分别通过模拟事例和数据中的控制区域，通过四轻子质量模板形状获得了四轻子质量谱，分别用于观测量的微分测量的每个单元，如参考文献中所述 [35]。

这些测量结果对于检验标准模型希格斯相关的预测至关重要，并且可以用于约束许多超标准模型理论的相空间。将所有积分和微分测量值与基于标准模型的理论预测值进行比较，发现在误差范围内是一致的。

当累积了更多的数据，我们可以获得更精确的结果，即更精确的希格斯玻色子性质测量，从而有可能发现与标准模型预测的任何可能偏差，这也是通往新物理学的大门。即使是基于全部第二运行阶段的数据，我们还可以研究更为细致的微分截面，以及更多的观测量。另外当前测量是基于POWHEG和NNLOPS的预测，我们可以研究更多的理论预测，并与微分截面测量值进行比较。此外，还可以进一步基于微分观测量进行潜在的理论解释。

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

Abstract

Chinese Academy of Sciences

Doctor of Science

The measurement of Higgs production cross section in the four-lepton final state at CMS

by Tahir JAVAID

The measurement of properties of Higgs boson is presented in its four-lepton decay channel. The fiducial cross sections (integrated and differential) of the Higgs boson production via $H \rightarrow 4\ell$ decays ($\ell = e, \mu$) are measured using the data collected with CMS detector in pp collisions at $\sqrt{s} = 13$ TeV. The measurements use Higgs mass $m_H = 125.09$ GeV and correspond to an integrated luminosity of 137.1 fb^{-1} at 13 TeV (Full Run 2 data). These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

In order to make all the measurements independent of how Higgs is produced, all the cross sections are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.}) \text{ fb}$ at 13 TeV. The differential fiducial cross sections are presented as a function of several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. These measurements are important to test the Standard Model predictions in the Higgs sector and can be used to constrain phase spaces of many theories beyond the Standard Model.

All the integrated and differential measurements are compared with the theoretical predictions based on the Standard Model and found to be consistent within their uncertainties.

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- 1.1 Summary of the CSC electronics boards that will be upgraded during LS2 and LS3. Boards above the horizontal line are placed on-chamber, while those below the line occur in off-chamber crates. Of the 540 new DCFEBv2s, 504 will be installed in the ME1/1 chambers, while the rest plus the DCFEBs currently in ME1/1 will be installed in ME2/1. The ALCTs in stations 2, 3, and 4 will be upgraded during LS2 while those in station 1 will be upgraded in LS3. The ALCTs in ME1/3 do not need upgrading because they will be replaced with the ALCT-S6 currently installed in ME1/1. 8

Chapter 1

The Cathode Strip Chambers Upgrades after Run 2 operations

Higher luminosity at LHC will increase the sensitivity for the physics analyses particularly for the new physics phenomena. The ‘Phase-1’ data taking period will end in the year 2023 providing data volume upto 300 fb^{-1} , followed by a shutdown for the HL-LHC upgrade ending in 2026. The high luminosity data taking period with the upgraded LHC is called ‘Phase-2’ and is expected to last from 2026 till 2038. The HL-LHC will increase the projected integrated luminosity of 300 fb^{-1} for ‘Phase-1’, the current LHC period, by an order of magnitude to 3000 fb^{-1} in the coming two decades, for both the CMS and ATLAS detectors. Brief description has already been given in Section ??.

Besides the planned upgrades for other sub-detectors of CMS, CSCs of its muon system will undergo several upgrades for HL-LHC.

1.1 Main CSC upgrades for High Luminosity (HL)-LHC

Although CSCs are designed to give satisfactory performance throughout the HL-LHC operations, there are several phase 2 upgrades planned.

1.1.1 Electronics

The combination of increased CSC occupancy plus larger L1 trigger rates and latency requirements in Phase 2 will cause the pipeline storage of the front-end electronics (CFEBs and ALCTs) in certain rings to fill up and lead to unacceptable dead time. Additionally, as a result of the higher CMS readout rates, the output bandwidth of the data acquisition boards (ODMBs and DDU) will have to be expanded. The increased occupancy in the chambers will necessitate the upgrade of the high voltage system. Some CSC read-out electronic boards will need to be replaced in order to handle the more stringent trigger requirements. In general, the new boards will cope with higher L1 trigger rates and chamber occupancies by employing high-speed output optical links and faster processors.

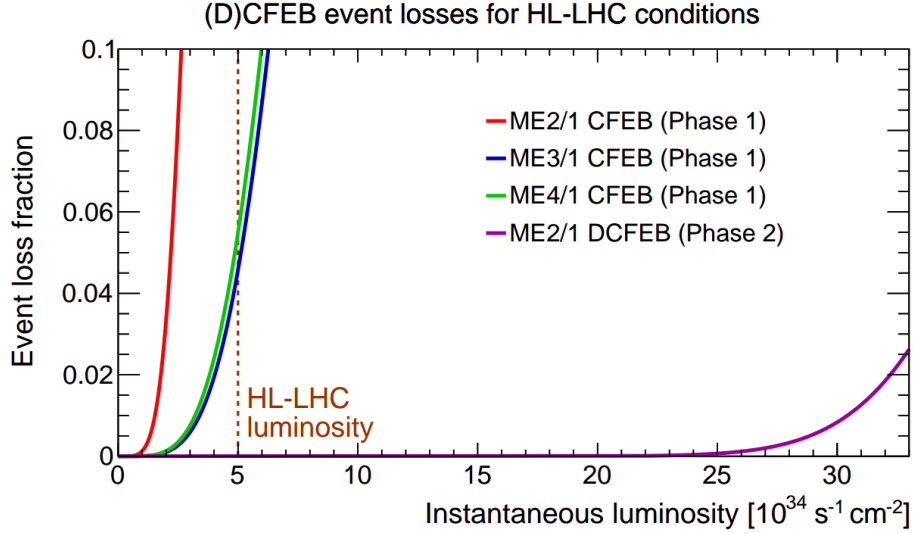


FIGURE 1.1: CFEB (Phase-1) and DCFEB (Phase-2) event loss fractions for HL-LHC conditions in the inner CSC rings ME2/1, ME3/1, and ME4/1 as a function of instantaneous luminosity. The design HL-LHC luminosity is marked by the dashed brown vertical line.

Cathode front-end boards

Existing cathode front-end boards (CFEBs) on all CSC rings except ME1/1 are based on analog technology of custom switched-capacitor array (SCA) chips that sample at 50 ns intervals to store the signal waveforms produced on the cathode strips when muon pass through them. Upon receiving a L1 accept signal, CFEBs set aside and digitize the charge measurements around the L1 time and the digital output is transferred to the data acquisition mother boards (DMBs). In parallel, fast trigger information is sent to the trigger mother boards (TMBs). These CFEBs are used on all CSCs except for those in the ME1/1 ring. This ring was equipped with digital cathode front-end boards (DCFEBs) in 2013-2014 to improve rate capability. In the ME2/1, ME3/1, and ME4/1 rings, which are also close to the beam, the HL-LHC trigger latency of 12.5 ms and the L1 trigger rate of 750 kHz will overwrite capacitor cells in the SCAs causing large readout inefficiency. Figure 1.1 shows the average event loss fraction for those stations as a function of the instantaneous luminosity.

The SCA overwriting problem can be addressed by replacing the 540 CFEBs in ME2/1, ME3/1, ME4/1 with DCFEBs, together with the replacement of the 108 TMBs servicing those chambers by OTMBs. New DCFEBs are being designed with an option that will allow for the remote programming of their FPGAs. As described in Section 4.5, these new boards will be installed in the ME1/1 chambers, thus mitigating any risks associated with their longevity, and the DCFEBs currently in the ME1/1 ring will be moved to chambers in the ME234/1 rings, which are subject to much lower levels of radiation. Top side of one of the DCFEBs produced for the ME1/1 upgrade having dimensions about $300\text{mm} \times 175\text{mm}$ with

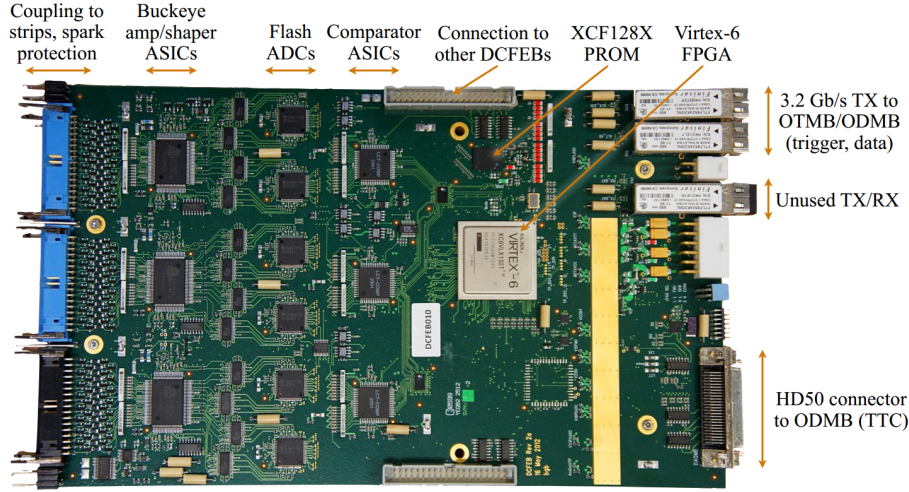


FIGURE 1.2: One of the DCFEB used for ME1/1.

chips mounted on both sides can be seen in the Figure 1.2. Only minor modifications with respect to the LS1 DCFEB design are adopted to develop new DCFEBv2s which are being installed during LS2.

Anode local charged track boards

The anode local charged track boards (ALCTs) detect the charge deposits on the CSC anode wires, form the corresponding trigger primitives, and send all this information to the DAQ upon receiving a L1 accept signal. With an increased L1 trigger latency time, these boards will not have sufficient FPGA memory resources to keep all the raw hit information. Moreover, the copper cable used in the

ALCTs to send the anode data to the TMBs does not provide enough output bandwidth for the boards in the rings closest to the beamline (ME1/1, ME2/1, ME3/1 and ME4/1) which will be insufficient for the expected HL-LHC data rates. Thus, all ALCT mezzanine boards will need to be replaced as well, except those in ME1/1 and ME4/2 chambers that were upgraded during LS1. In addition, the radiation hardness of the ALCT boards currently in ME1/1 is a concern, so they will be moved to ME3/1 and new radiation-hard ALCTs will be installed in ME1/1.

These problems are solved by replacing the mezzanine cards in the affected chambers by new boards based on the design of the ALCT-S6 developed for the LS1 upgrade (Fig. 4.11). The Spartan-6 FPGAs that will power the upgraded cards (XC6SLX100-3FVG676C and XC6SLX150T-3FVG900C) have 9'Ä12 times more memory than the currently used Virtex-E FPGAs, which allows for the pipeline storage to be made deep enough to satisfy the HL-LHC latency requirements. New boards are based on the design of the ALCT-S6 developed for the LS1 upgrade as shown in the Figure 1.3.

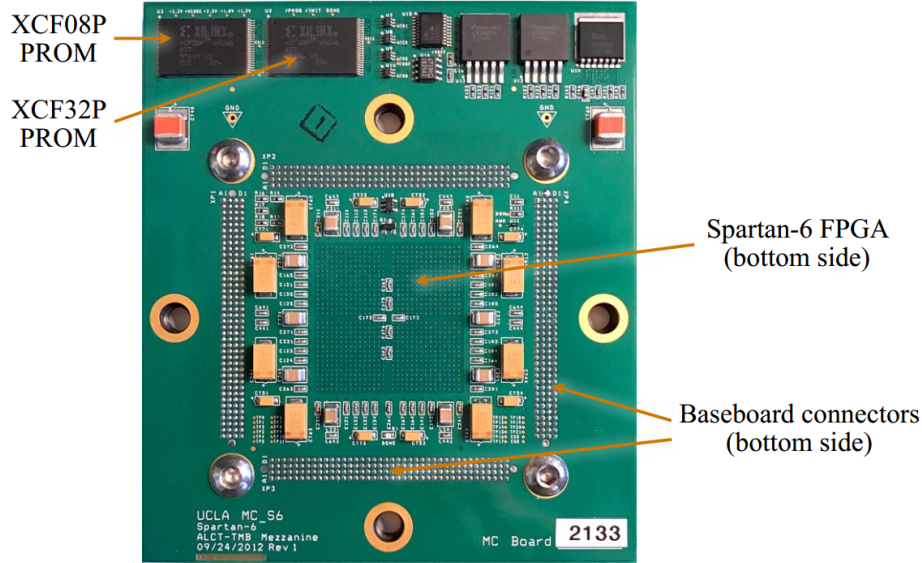


FIGURE 1.3: Top side of one of the Spartan-6 ALCT mezzanine boards produced for the ME1/1 upgrade. The upgraded ALCT-LX150T/LX100 mezzanine boards will be around $104\text{mm} \times 140\text{mm}$ to accommodate the new optical components.

TMB Upgrade

The relocated DCFEBs and the new DCFEBv2s in the ME234/1 rings will transmit trigger primitive data to the peripheral crates via 3.2 Gb/s optical links. Since the currently installed TMBs are designed to work with CFEBs, they have no optical receivers to accommodate the optical data link from the DCFEBv2s. This necessitates building 108 OTMBv2 mezzanine boards and 108 OTMBv2 main boards in LS2 for the ME2/1, ME3/1, and ME4/1 rings. An OTMB mezzanine board produced for the ME1/1 upgrade is shown in the Figure 1.4.

The OTMBs were designed and built during LS1 for the ME1/1 upgrade. The mezzanine boards employ a Reflex photonics Snap12 transceiver featuring 12 optical input receivers as shown in Fig. 4.12. This mezzanine board plugs into a VME OTMB baseboard. The boards also have a relatively modern Xilinx Virtex-6 FPGA with built-in high speed serial transceivers. The use of an upgraded FPGA provides the additional advantage of increased logic resources to improve the trigger algorithms in this high rate region. The Snap12 receivers are now obsolete so they will be replaced by another MTP12- type set of 12 receivers.

DMB and ODMB upgrades

The 1.6 Gb/s output link in the DMBs and ODMBs will not be able to handle the data rates in the inner CSC rings expected under HL-LHC running conditions. Two similar boards based on the ODMB design developed for the LS1 upgrade will increase the output bandwidth of the affected stations: the 72 ODMBs in ME1/1 will

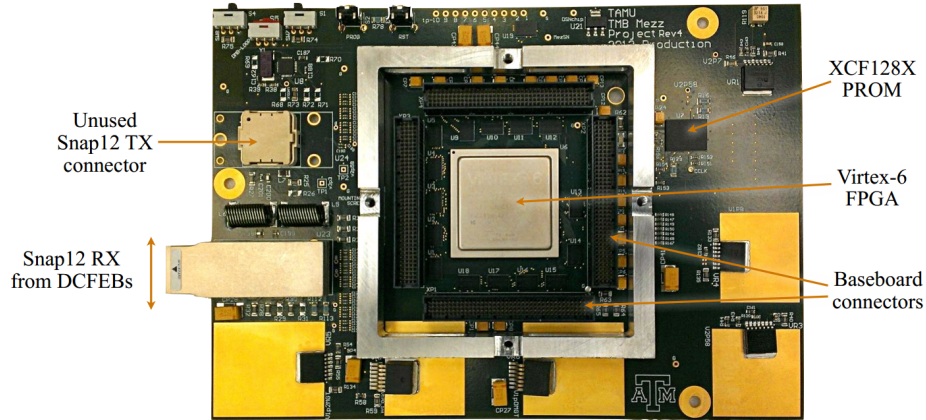


FIGURE 1.4: Bottom side of one of the OTMB mezzanine boards produced for the ME1/1 upgrade. The OTMB mezzanine is about $190\text{mm} \times 134\text{mm}$ in size.

be replaced by ODMB7s that will connect to the seven DCFEBv2s in each ME1/1 CSC via the patch panel shown in Section ?? Figure. ?? (right), and the 108 DMBs in ME234/1 will be replaced by ODMB5s that will connect directly to the five DCFEB(v2)s in those CSCs as shown in the right of the same Figure. New optical fibers connecting the DCFEB(v2)s in ME234/1 with the corresponding ODMB5s will be laid out.

The main difference between the designs of the ODMB5 and the ODMB7 lies in the front panel. For the LS1 upgrade of ME1/1, the five CFEBs on each CSC module were replaced by seven DCFEBs. Since there was not enough space on the front panel of the ODMB to fit seven HD50 connectors, a patch panel was used to reroute and fan out the signals coming from two 50-pin Skewclear cables connected to the ODMB to the seven cables going to the DCFEBs. The front panel of the ODMB7s will be largely the same as that in the current ODMBs, shown in Figure 1.5. The design of the backplane connections, the clock and power distribution, and the FPGA and PROM will be the same for the ODMB5 and ODMB7, and will be very similar to the design of those parts in the existing ODMB. The main modification with respect to the ODMB corresponds to the upgrade of the FPGA from a Virtex-6 to the largest and highest speed grade Artix-7 model. The ODMB5 and ODMB7 will retain the upstream optical link with the FED board.

FED upgrade

For Phase 2, the CSC FED system will be upgraded to handle the 900 optical links at speeds up to 6.4 Gb/s (current (O)DMB output links running at 1.6 Gb/s), delivering a total data rate of 598 Gb/s.

The proposed architecture for the new CSC FED boards will likely be based on the ATCA platform, the common CMS format for Phase 2. The trigger group is designing new generic ATCA boards with different FPGA sizes. These boards

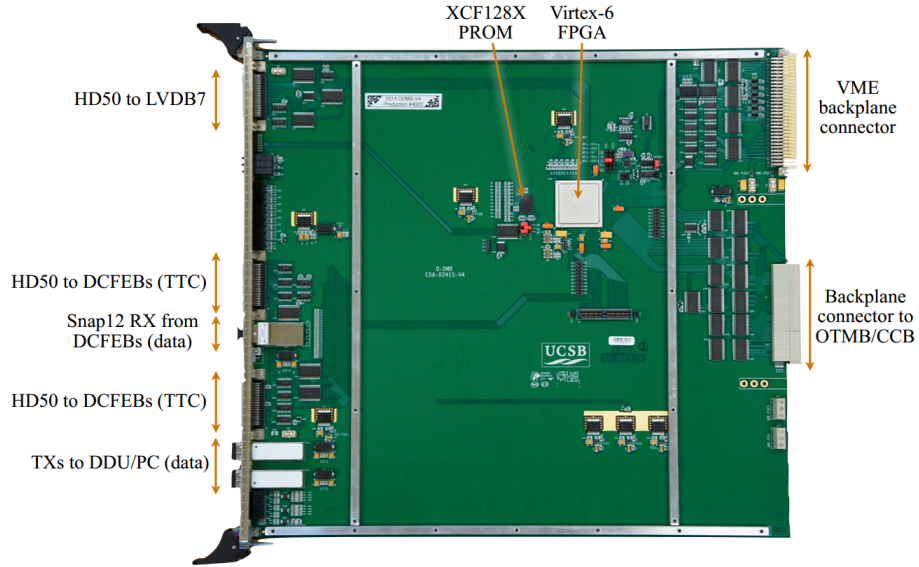


FIGURE 1.5: Top side of one of the ODMBs produced for the ME1/1 upgrade. The ODMB is a 9U VME board, $400\text{mm} \times 367\text{mm}$ in size. The upgraded ODMB7 will be very similar to the ODMB, while the ODMB5 will have a different front panel configuration with copper connections to the five DCFEB(v2)s.

would each have 80 to 104 optical links with speeds up to 16 or 30 Gb/s, and also include a Zynq processor card and large memories that could handle the remote programming of the DCFEB and ALCT FPGAs, exactly as needed for the CSC DAQ. A total of twelve of these boards would be able to handle the CSC data rates expected at the HL-LHC.

Low and high voltage systems

The low voltage (LV) and high voltage (HV) systems will also need to be upgraded to ensure the reliable operation of the CSCs at the HL-LHC. The new electronics will consume more power, which will require the redesign of the LV system with higher maximum currents and appropriate voltage levels. Higher background rates and the need for a more accurate monitoring of CSC currents will also imply an incremental upgrade for the present HV system.

Figure 1.6 shows a simplified overview of the CSC electronics upgrade. Given that the installation schedule in LS3 is driven by the replacement of various detectors elements in the barrel region, most of the CSC replacements that require chamber access will be performed ahead of that, during LS2. All the CFEBs in the radially innermost CSCs will be equipped with slightly modified DCFEBs to cope with the HL-LHC requirements. The DCFEBs use 12-bit 20 MHz flash ADCs and powerful Virtex-6 FPGAs with large internal memory resources. The resulting digital pipeline can hold up to 700 events, which leads to negligible dead time at the HL-LHC.

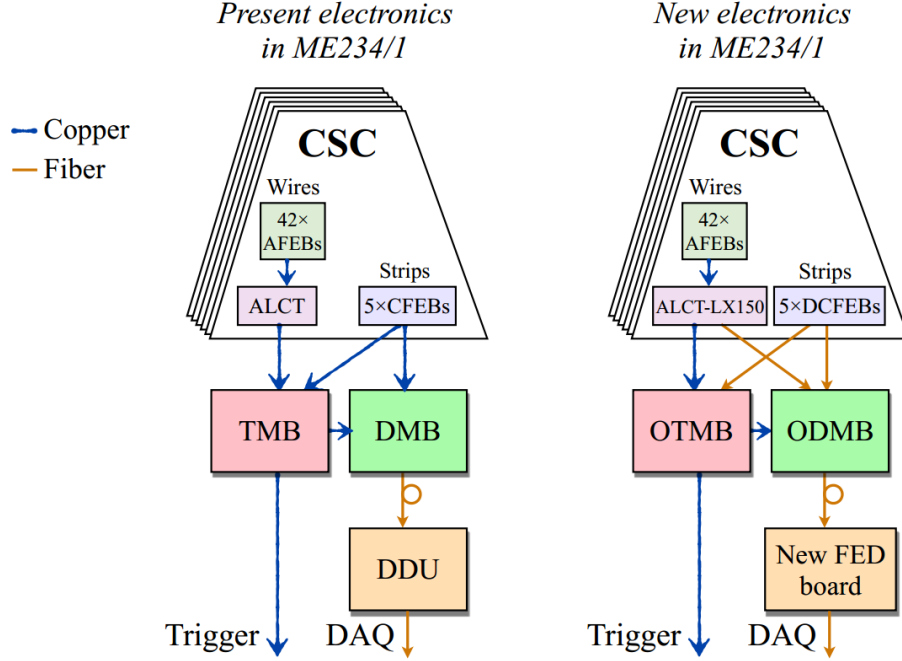


FIGURE 1.6: Simplified schematic of the current (left) and of the new (right) CSC electronics for the inner rings formed by ME2/1, ME3/1 and ME4/1. The orange lines mark optical data transfer, the blue lines represent copper cables..

Additionally, radiation tests of the XCF128X PROMs employed in the current DCFEBs show that they may not withstand the radiation doses expected at the HL-LHC. Thus, the new DCFEBs are being designed with an option that employs the CERN-designed GBTx ASIC to allow for the remote programming of their FPGAs. These new boards will be installed in the ME1/1 chambers, thus mitigating any risks associated with the longevity of their PROMs, and the DCFEBs currently in the ME1/1 ring will be moved to other chambers in the ME2/1, ME3/1, and ME4/1 rings subject to much lower levels of radiation.

The DCFEBs send the trigger data via 3.2 Gb/s optical links, so the TMBs servicing the inner rings will be upgraded to optical TMBs (OTMBs). The low voltage distribution boards (LVDB) that supply power to the DCFEBs will be upgraded as well. The mezzanine boards for the ALCTs in both the inner and outer CSC rings (396 chambers in total), not including the already upgraded ME1/1 and ME4/2 rings, will be upgraded with more powerful Spartan-6 FPGAs. The inner ring boards will also incorporate optical links to cope with the HL-LHC data rates that will provide an output bandwidth of up to 6.4 Gb/s, ten times higher than previously. Finally, the boards to be installed in ME1/1 will be outfitted with the same remote FPGA programming option that the new DCFEBs will have, thus mitigating any risks associated with the aging of the PROMs in that ring. Similarly, the DMBs in the inner rings will be upgraded with new optical DMBs (ODMBs) to handle the HL-LHC data rates. These boards will have output bandwidths of up

to 25.6 Gb/s, achieved with multiple 6.4 Gb/s links. The back-end electronics collecting all the CSC data will be upgraded accordingly. Altogether, a total of 1044 on-chamber electronics boards and 302 trigger and DAQ modules will be replaced in the CSC system.

During ongoing long shutdown 2 (LS2), there several upgrades are in process for upcoming physics programs under high luminosity LHC. CSC will undergo upgrade of readout system of ME1/1, ME2/1, ME3/1, and ME4/1. Total 180 chambers will be refurbished to replace CFEBs with upgraded DCFEBs and ALCT mezzanine boards. Also for ME2/1, ME3/1 and ME4/1, LVDBs will also be upgraded.

The detailed description of the upgrade plan of the involved electronics is given in the Table 1.1.

Board →	Number	Where	When	Main reasons for upgrades
DCFEBv2	540	ME12/1	LS2	Latency, rate, rad-hard requirements
LVDB5	108	ME234/1	LS2	Power levels of DCFEBv2s
ALCT-LX100	288	ME1/1, ME123/2	LS2/LS3	Latency, rate, rad-hard requirements
ALCT-LX150T	108	ME234/1	LS2	Latency and rate requirements
OTMBv2	108	ME234/1	LS2	Receive optical link from DCFEBv2s
ODMB7	72	ME1/1	LS3	Increased DAQ output bandwidth
ODMB5	108	ME234/1	LS3	Increased DAQ output bandwidth
HV master	40	ME1234/1	LS3	Increased current due to higher occupancy
HV distribution	12	ME1234/1	LS3	Increased current due to higher occupancy
FED	14	USC	LS3	Increased data volume, number of links

TABLE 1.1: Summary of the CSC electronics boards that will be upgraded during LS2 and LS3. Boards above the horizontal line are placed on-chamber, while those below the line occur in off-chamber crates. Of the 540 new DCFEBv2s, 504 will be installed in the ME1/1 chambers, while the rest plus the DCFEBs currently in ME1/1 will be installed in ME2/1. The ALCTs in stations 2, 3, and 4 will be upgraded during LS2 while those in station 1 will be upgraded in LS3. The ALCTs in ME1/3 do not need upgrading because they will be replaced with the ALCT-S6 currently installed in ME1/1.

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